The Alignment Tax on Humor is a Safety–Fine-Tuning Tax, Not an RLHF Tax

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Abstract

A common worry about language-model safety training is that it imposes an *alignment tax* on creativity: by steering models toward safe, predictable text it may flatten the statistical properties that make outputs surprising, clever, and funny. We test this directly on humor using Llama 3.2 1B across three alignment tiers (base, RLHF-instruct, instruct + LoRA safety fine-tune), three judge-independent distributional metrics (punchline surprisal, pre-punchline entropy, setup–punchline embedding distance), an LLM-as-judge rubric, and linear probes. We find four things. (i) The humor tax is *non-monotonic*: RLHF does not tax humor—instruct jokes are rated as funny as a human corpus (6.77 vs. 6.71/10)—whereas the safety fine-tune collapses humor ($\rightarrow 3.24$), driven by a 74% refusal rate on benign joke prompts. (ii) Alignment *widens* rather than compresses all three metrics at every stage (entropy $+0.44$ instruct$\rightarrow$safety), the opposite of the compression hypothesis, and this replicates cross-lingually on Chinese Oogiri and Ruozhiba humor (d 0.43–0.83). (iii) The classic inverted-U between distributional "surprise" and rated humor is not supported ($R^2 \leq 0.09$), and raising temperature monotonically *lowers* humor. (iv) The tax is *asymmetric*: fine-tuning for humor does not erode AdvBench/HarmBench safety (refusal $\sim 93\%$, tied with the baseline). Probes show humor *understanding* survives in the safety model even as *production* collapses—the tax falls on production, not comprehension. We also document that a local 1B judge scores only $\sim 18\%$ of generated jokes validly, biasing any kept subset.

1 Introduction

RLHF and downstream safety fine-tuning are standard steps in shipping a language model (Ouyang et al., 2022; Bai et al., 2022; Rafailov et al., 2023). A recurring concern is the *alignment tax*: that the optimization making a model helpful and harmless also degrades capabilities depending on exploration, surprise, or low-probability completions (Askell et al., 2021; Bai et al., 2022). Humor is a sharp test case. Computational accounts since Raskin (1985) and Attardo & Raskin (1991) center on *incongruity-resolution*: a joke builds an expectation and the punchline violates it (Suls, 1972). If safety training compresses outputs toward safe, high-probability text, it should sand down precisely that incongruity.

We make this measurable. We study Llama 3.2 1B (Dubey et al., 2024) at three alignment tiers—a base completion model, the RLHF instruct model, and an instruct model given an additional LoRA (Hu et al., 2022) safety fine-tune on AdvBench (Zou et al., 2023) and HarmBench (Mazeika et al., 2024) refusals—and at each tier ask both *how funny* its jokes are (LLM-as-judge (Zheng et al., 2023)) and *how its output distribution moves* on a fixed corpus of human jokes, via three judge-independent metrics: punchline surprisal, pre-punchline entropy, and setup–punchline embedding distance.

Our results overturn the simple "RLHF flattens humor" story. The tax is *non-monotonic in alignment*: RLHF produces the best joke-writer in our study (tied with the human corpus), and the tax appears only at safety fine-tuning, where it is dominated by *refusal* of benign prompts. Alignment does not *compress* the distribution at all—every metric *widens* at every stage, a pattern that replicates on Chinese humor. We find no inverted-U between distributional surprise and rated humor, and we show the tax is *asymmetric*: pushing toward more humor does not measurably erode safety. We also report a methodological caveat on local judges.

Contributions. (1) The humor tax is non-monotonic and a *safety-fine-tuning* effect, not an RLHF effect, and is largely a refusal (behavioral) effect (§3). (2) Alignment *widens*, not compresses, all three metrics at every tier, replicating cross-lingually on Oogiri and Ruozhiba (§4). (3) No inverted-U, and temperature monotonically lowers humor (§5). (4) The tax is *asymmetric* (humor FT keeps safety), while probes find humor understanding intact in the safety model (§6). (5) A local 1B judge is unusable on generated jokes ($\sim$18% valid), biasing any kept subset (§2).

2 Method

Models. All tiers share the Llama 3.2 1B architecture. **Base** is the pretrained completion model; **Instruct** is Meta's RLHF model (SFT + rejection sampling + DPO (Rafailov et al., 2023)); **Safety** is Instruct plus a LoRA refusal fine-tune (r=16, 3 epochs, $\eta = 2 \times 10^{-4}$, response-only loss masking) on 920 (harmful prompt, refusal) pairs from AdvBench and HarmBench, with seven refusal templates (loss $3.72 \rightarrow 0.34$).

Metrics. For each joke we compute three judge-free quantities. *Punchline surprisal*: mean negative log-probability of the punchline given the setup (high = unpredicted). *Pre-punchline entropy*: Shannon entropy of the next-token distribution at the end of the setup (high = many plausible continuations). *Setup–punchline embedding distance*: cosine distance between sentence embeddings of setup and punchline (high = larger semantic leap). These operationalize incongruity and track distributional movement on a *fixed* corpus, independent of any judge.

Data, generation, judge, probes. The human corpus is 2,000 jokes (1,000 each from r/Jokes (Weller & Seppi, 2020) and a one-liner set; capped for cost). Generated jokes come from a 7-config decoding sweep (temperature 0.3–1.6 $\times$ narrow/wide nucleus), 50/config/model, yielding 159/350/350 valid base/instruct/safety jokes. We score jokes with Gemini 2.5 Flash (structured outputs) on a 1–10 rubric (*unexpectedness*, *cleverness*, *amusement*, *overall*), with Gemini 2.5 Pro on a subset for an agreement check. Probes are linear classifiers on hidden states predicting whether an explanation correctly accounts for a joke, on Chumor (He et al., 2024) with 5-fold `GroupKFold` (majority baseline 56.5%).

A local judge is unusable on generated jokes. A local Llama-1B judge emits valid structured scores for only 151/859 ($\sim$18%) of generated jokes, with failures concentrated on the noisiest (high-temperature, safety) outputs—so any analysis on the kept subset is biased toward easy cases. A single API judge over both corpus and generated jokes is 99.7% valid (2,851/2,859) and removes the original two-judge confound (the corpus had been judged locally while generated jokes were never judged). On a 110-joke subset, Flash–Pro agreement is moderate (mean $|\Delta\text{overall}| = 1.52$, within-1 66%); we therefore treat absolute scores as judge-dependent ($\pm$1.5) and rely on relative comparisons within one judge.

3 The humor tax is non-monotonic and safety-driven

Humor does not decline with alignment; it rises with RLHF and then collapses with safety fine-tuning (Table 1, Figure 1). The instruct model (6.77 overall) is tied with the human corpus (6.71) and is the best joke-writer in the study. The tax appears only at the safety tier (3.24) and is largely *refusal-driven*: 258/350 (74%) of safety-model outputs are refusals or disclaimers ("can't help with that") to entirely benign joke prompts; refusal-like outputs average 3.03, and even the non-refusals reach only 3.83. A base *completion* model barely follows "write a joke," so its low 3.67 is partly off-format output; the trustworthy contrast is therefore **instruct** $\rightarrow$ **safety**, where both models follow the instruction but only safety collapses.

Table 1: Mean Gemini-judge humor by tier vs. the human corpus (1–10, higher funnier). Non-monotonic: RLHF matches the corpus; safety fine-tuning collapses humor.

Tier	n	overall	unexpectedness	cleverness	amusement
Corpus (human jokes)	1993	**6.71**	7.00	6.50	6.71
base (no alignment)	159	3.67	4.22	3.77	3.37
instruct (RLHF)	349	**6.77**	6.25	7.40	6.63
safety (RLHF + safety FT)	350	3.24	3.38	3.49	2.93

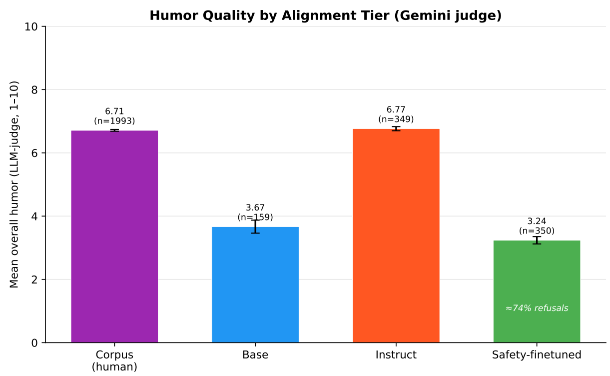

Figure 1: Humor by alignment tier (Gemini judge). RLHF matches the human corpus; safety fine-tuning collapses humor, with $\sim$74% of its outputs refusals to benign joke prompts.

4 Alignment widens the distribution—it does not compress it

If safety training compressed outputs, surprisal, entropy, and embedding distance should *shrink*. They do the opposite. Assigning each metric to the *same* 2,000 human jokes (Table 2), every metric *increases* at every stage. The standout is entropy: the safety fine-tune raises pre-punchline entropy $+0.44$ over instruct—the safety model is markedly more *uncertain when reading* human jokes even as it largely *refuses to write* them (§3). It is simultaneously worse at modeling humor and unwilling to produce it.

Cross-lingual replication. The widening is not an artifact of English Reddit humor. We fetch the two Chinese datasets named in our proposal—Oogiri (crowd-sourced odai$\rightarrow$boke pairs from CLoT (Zhong et al., 2024)) and Ruozhiba (absurd-logic Q&A (Ruozhiba, 2024)), 500 each—and run identical metric extraction (Table 3). In both, base$\rightarrow$instruct widens all three metrics (d 0.43–0.83) and base$\rightarrow$safety widens further, mirroring English. As a judge sanity check, Oogiri scores 6.76 overall (as funny as the English corpus) while Ruozhiba scores 2.95—the judge cleanly separates crowd-humor from absurd-logic Q&A.

5 No inverted-U, and temperature lowers humor

Incongruity theory predicts humor should peak at *moderate* surprise, an inverted-U between each metric and rated humor. We do not find it. Fitting humor $= az^2 + bz + c$ on the corpus (12 metric$\times$dimension combinations), no combination shows a significant negative quadratic and $R^2 \leq 0.089$ everywhere. On *generated* instruct jokes the metrics relate to humor more strongly (R^2 up to 0.32) but monotonically, not as a peak (Figure 2). A significant inverted-U appears only in the safety tier, best explained as an artifact of the 74%-refusal mixture. The temperature sweep agrees: as temperature rises $0.3 \rightarrow 1.6$, surprisal and entropy climb but instruct humor *declines monotonically* ($7.22 \rightarrow 5.76$)—funniest at low temperature / narrow nucleus, with no peak (Figure 3).

Table 2: Distributional shift across tiers on a fixed human corpus (judge-independent). Every metric *widens* at every stage—opposite of the compression hypothesis. $^{***}p < 0.001$, $^{*}p < 0.05$, ns = not significant; Cohen's $d \in [0.05, 0.45]$.

Metric	base→instruct	instruct→safety	base→safety
punchline surprisal	$+0.339^{***}$	$+0.092^{*}$	$+0.431^{***}$
pre-punchline entropy	$+0.210$ (ns)	$+0.438^{***}$	$+0.648^{***}$
embedding distance	$+0.035^{***}$	$+0.008$ (ns)	$+0.043^{***}$

Table 3: Cross-lingual replication on Chinese humor. base→instruct ($b{\to}i$) and base→safety ($b{\to}s$) widen every metric in both datasets (Cohen's d).

Source	Metric	base	instruct	safety	$b{\to}i$ d	$b{\to}s$ d
Oogiri	surprisal	4.50	5.32	5.48	$+0.61$	$+0.73$
	entropy	8.66	9.41	9.35	$+0.74$	$+0.70$
	distance	0.158	0.211	0.209	$+0.74$	$+0.70$
Ruozhiba	surprisal	2.34	2.55	2.72	$+0.43$	$+0.75$
	entropy	7.19	8.32	8.59	$+0.50$	$+0.62$
	distance	0.101	0.135	0.123	$+0.83$	$+0.56$

6 The tax is asymmetric, and understanding survives

No reverse tax. We test the mirror question—does pushing toward *more* humor erode safety?—with a net-new LoRA SFT of the instruct model on 1,257 corpus-joke completions (*Instruct-humor*) plus an AdvBench/HarmBench evaluation harness (greedy generation + standard refusal-substring heuristic). On 120 held-out harmful prompts (Table 4), humor fine-tuning leaves refusal essentially unchanged (93.3% vs. baseline 90.0%), nowhere near collapse. Qualitatively the asymmetry is stronger: the baseline's 12 non-refusals are mostly genuine soft-compliance, whereas the humor model's 8 are *degenerate deflection*—it falls back to its joke distribution (e.g. "I'm a little teapot" for requested lyrics) rather than complying. Safety is cheap to keep while adding humor; humor is expensive to keep while adding safety. (The refusal heuristic is a substring match over $n = 120$, so $\pm$ a few prompts is noise; note model text must be Unicode-normalized first—curly apostrophes in "can't" initially mis-scored 10 refusals and inverted the headline.)

Understanding survives, production does not. Linear probes (Table 5) show all tiers carry a real but weak humor-understanding signal (AUC 0.61–0.62, above the 56.5% baseline). RLHF pushes humor features toward the output layers (peak layer $11 \to 16$). Crucially, the safety model keeps the signal late (layer 15) and intact (AUC $0.618 \approx$ instruct's 0.619), not flattened. So the same fine-tune that collapses joke *generation* (§3) and widens reading-time entropy (§4) leaves humor *comprehension* untouched: the tax falls on production, not understanding.

7 Discussion and conclusion

The popular framing—"safety training makes models blander"—is wrong in its mechanism here. There is a real, large humor tax, but it is (i) located at *safety fine-tuning*, not RLHF; (ii) driven by *refusal* of benign requests, not lower-incongruity jokes; and (iii) accompanied by a *widening*, not narrowing, of the output distribution that replicates cross-lingually. A model that refuses to tell a joke and is more uncertain reading one is failing through over-triggered refusal and degraded language modeling, not safe-and-boring compression. The mitigation lever is therefore refusal calibration on benign creative prompts, not "turning down safety"—and the asymmetry (humor FT keeps safety) is an encouraging sign that creative capability can be added to aligned models cheaply.

Limitations. Single model scale (1B); effects may differ at 7B+. Absolute judge scores are judge-dependent (± 1.5); we rely on within-judge ordering. The base tier is confounded by weak instruction-following, so we anchor claims on instruct→safety. The safety degradation is refusal-

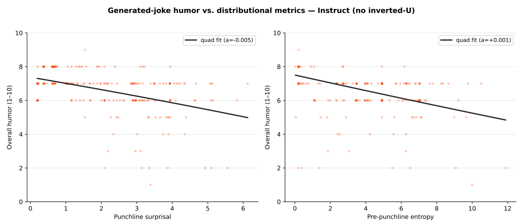

Figure 2: Humor vs. metrics for generated instruct jokes, with quadratic fits. No significant negative quadratic; the relationship is monotonic.

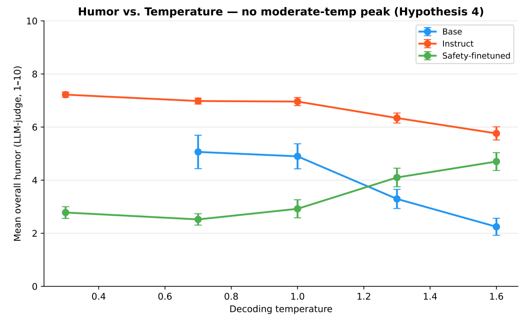

Figure 3: Humor vs. temperature (instruct). Surprisal/entropy rise but humor falls monotonically ($7.2 \to 5.8$)—no inverted-U.

Table 4: Reverse direction: refusal on 120 harmful prompts (higher = safer; ASR = 1−refusal). Humor FT does *not* erode safety.

Model	refusal	ASR
Instruct (baseline)	0.900	0.100
Instruct-**humor**	**0.933**	0.067
Instruct-safety	1.000	0.000

Table 5: Linear-probe humor understanding on Chumor (majority 56.5%). RLHF moves the signal late; safety keeps it late and intact.

Model	Best layer	AUC
Base	11 (mid)	0.608
Instruct	16 (final)	0.619
Instruct-safety	15 (near-final)	0.618

dominated (partly behavioral), and the reverse-tax eval uses a substring heuristic over 120 prompts. A blind 120-joke human-rating subset is prepared to validate the judge but not yet hand-rated.

Conclusion. Across distributional analysis, LLM-as-judge scoring, cross-lingual replication, a reverse fine-tuning experiment, and probing, the humor alignment tax in Llama 3.2 1B is non-monotonic, asymmetric, and mechanistically the opposite of the usual story: RLHF improves humor, safety fine-tuning collapses it through refusal, alignment widens rather than compresses the distribution, there is no inverted-U or temperature sweet spot, and humor understanding survives even when production does not.

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A Additional details

Decoding sweep. The seven configurations are low_temp (T=0.3), mid_temp (T=0.7), narrow_nucleus (T=1.0, p=0.5), default (T=1.0), wide_nucleus (T=1.0, p=0.9), high_temp (T=1.3), very_high_temp (T=1.6); instruct overall humor is $7.22, 6.98, 7.26, 6.96, 6.84, 6.34, 5.76$ respectively. Safety fine-tuning: LoRA r=16, 3 epochs, $\eta = 2 \times 10^{-4}$, response-only masking; loss $3.72 \rightarrow 0.34$, token accuracy $\approx 90\%$. Code, judgments, metrics, probe outputs, and per-prompt safety responses are saved as artifacts in the project repository.

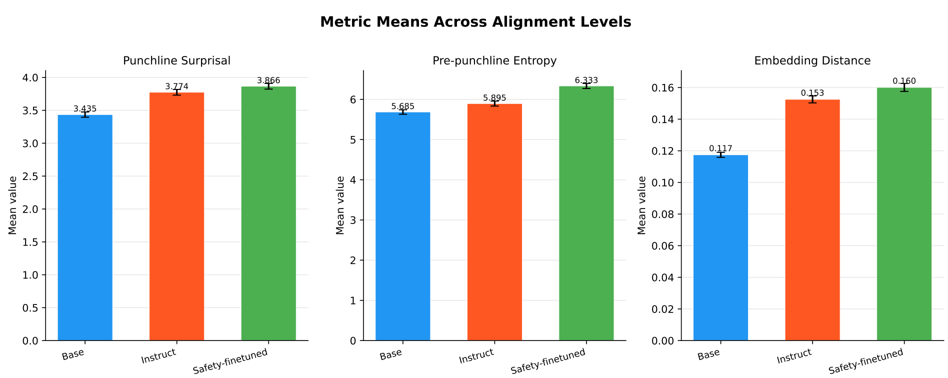

Figure 4: Per-tier means of the three distributional metrics on the fixed human corpus (underlying Table 2).